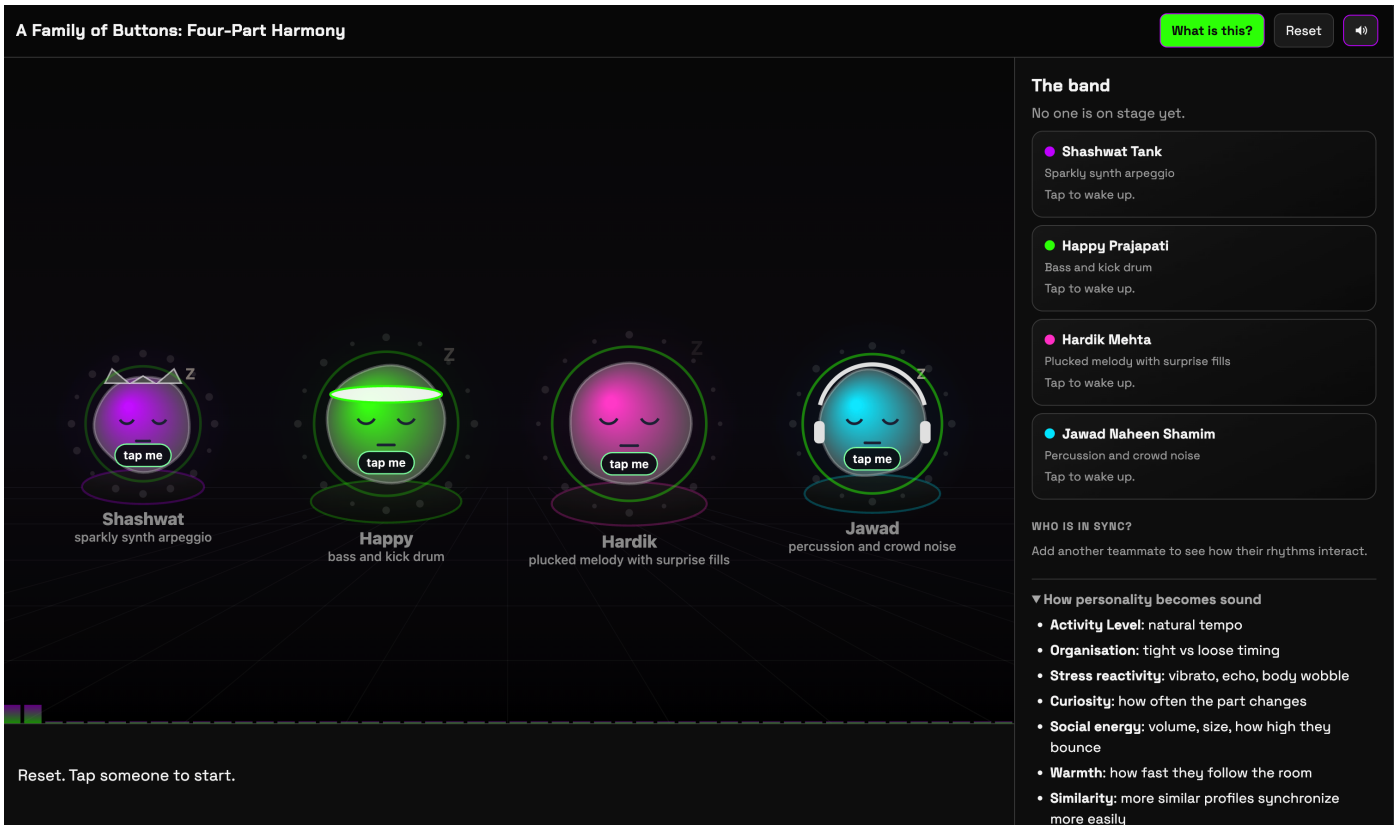
A Family of Buttons

Four-Part Harmony

Link to the Application: https://fourpartharmony.netlify.app

CPSC 581| Group Project | Tutorial 2 Group 5

Team members
Hardik Mehta | Shashwat Tank | Happy Prajapati | Jawad Naheen Shamim



Final application interface

Concept
Personal interests make each teammate recognizable, personality influences how each teammate behaves, and interaction reveals how the team works together.

1. Project overview

Four-Part Harmony is an interactive web application built around exactly one teammate button for each group member. Tapping a teammate wakes them up as a musician. Their instrument is inspired by their hobbies, while their musical and visual behavior is influenced by Big Five Personality (ITP survey). When several teammates are active, the application reveals team dynamics through rhythmic interaction, visual connections, and synchronization. Users can also move teammates closer together to strengthen their interaction.

The main design challenge was to represent four distinct people without turning the application into four isolated profiles. We therefore treated individual identity and team dynamics as two connected goals; each button should communicate who a teammate is and combinations of buttons should reveal how teammates relate to one another.

2. Design goals

- Make every teammate recognizable through a combination of personality-driven behavior and a personal interest.
- Represent personality through interaction, sound, movement, and timing.
- Make similarities and differences between teammates visible when multiple buttons are active.
- Give the user meaningful control over team interaction instead of making the application a passive visualization.
- Keep the final interface understandable and visually focused, with supporting text secondary to the stage.

3. Design process at a glance

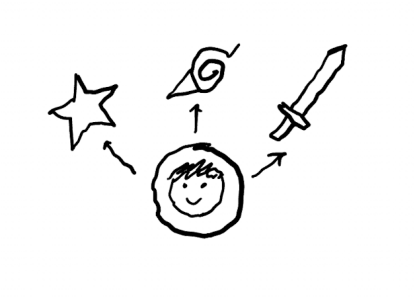


We used a divergent-convergent process. Early sketching explored several ways to represent individual teammates and a shared team. We then converged on the band metaphor because music could express both individual personality and relationships between people. The concept was implemented in stages, reviewed for clarity and usability, and refined through changes to synchronization, dragging feedback, information hierarchy, and visual identity.

4. Divergence - Exploring individual representations

We began with low-fidelity sketching because it allowed us to generate alternatives quickly without committing to implementation. The early concepts focused on recognizable interests and visual metaphors for each teammate. This stage intentionally prioritized breadth over polish.

SKETCH 1
Shashwat - anime concept



Tap to cycle through anime symbols, the symbols float around

SKETCH 2
Happy - gym concept



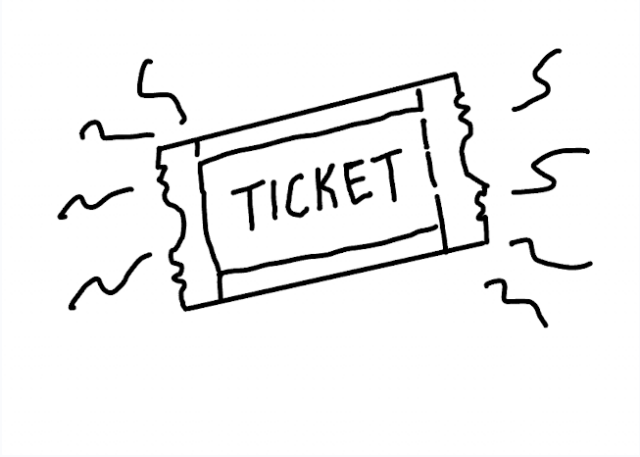
Tap to make Happy lift weights, increase size and show reps

SKETCH 3
Hardik - travel concept



Tap to open suitcase, items like plane ticket, map shows up

SKETCH 4
Jawad - events concept



Tap to show a ticket opens a which event Jawad is attending

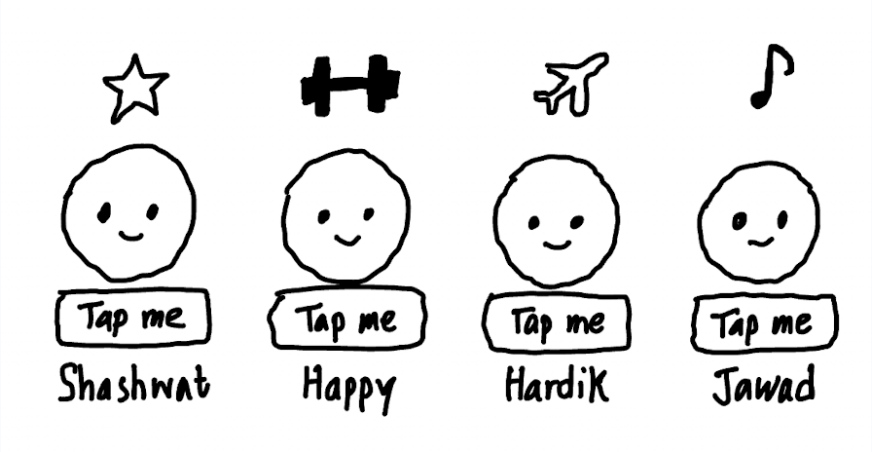
These concepts were useful for individual likeness, but they exposed a common weakness each idea worked mainly as a separate personal profile rather than the interactions between buttons to reveal team dynamics, so we needed a shared metaphor that could connect all four people.

5. Divergence - Moving from four individuals to one team

The next step was to explore a shared environment. Instead of asking only "What represents each person?", we asked "What happens when all four buttons exist in the same system?"

SKETCH 5

Shared room concept



Everyone occupies the same environment. Each character shows their interest.

5.1 Comparing candidate directions

We compared the directions against criteria taken from the project requirements: individual identity, team dynamics, interaction potential, novelty, clarity, and feasibility. This qualitative comparison helped us converge without treating visual preference as the only decision criterion.

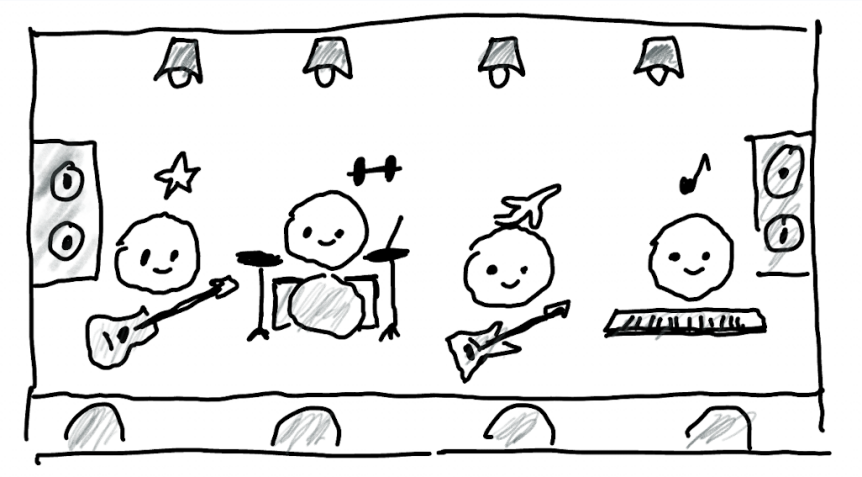
Candidate direction	Individual identity	Team dynamics	Main limitation / decision
Separate interest-based buttons	Strong	Limited	Recognizable people, but the buttons behaved like four separate profiles.
Shared room	Moderate	Moderate	Placed everyone together, but did not provide a strong metaphor for personality differences.
Reaction-field prototype	Moderate	Strong	Relationships were visible, but the interaction felt abstract and required explanation.
Music band	Strong	Strong	Sound, rhythm and synchronization could represent both individuality and relationships. Selected for further development.

The band concept was selected because it solved both sides of the design problem at once. Musical properties could represent how an individual behaves, while synchronization could represent what happens when multiple personalities interact. It also gave users a reason to activate several buttons together rather than treating each button independently.

6. Convergence - Choosing the band concept

SKETCH 6

Initial music band concept



The first band concept brought all four teammates into one shared stage and gave each person a distinct musical role.

6.1 Why a band?

We chose the band metaphor because personality could become behavior rather than a static description. Tempo, timing precision, variation, movement, musical presence, and responsiveness are all perceivable qualities. This allowed users to hear and see differences without first understanding the underlying personality survey.

The band also created a natural relationship metaphor- musicians can drift, listen, adapt, and synchronize. This gave us a way to express team dynamics through the behavior of the buttons themselves.

6.2 Personality-to-behavior mapping

Personality input	Application behavior	Design rationale
Activity Level	Natural tempo (BPM)	Tempo is a direct, perceivable expression of pace and activity.
Conscientiousness / organization	Tighter or looser timing	Rhythmic precision communicates structure without requiring a score display.
Emotionality / stress reactivity	Vibrato, echo and body wobble	A more reactive sound and movement makes the trait perceptible.
Openness / curiosity	How often the musical pattern changes	Variation represents experimentation and novelty.
Extraversion / social energy	Musical presence, size and bounce	Outward energy becomes visible and audible.
Agreeableness / warmth	How readily a rhythm follows others	The trait affects interpersonal behavior rather than only appearance.

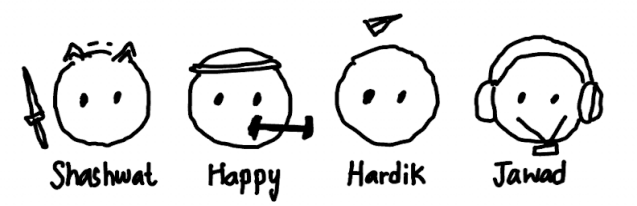
Identity vs Behavior

Hobbies are used mainly for recognizable likeness (travel, anime, gym, events). Personality data is used mainly for behavior. This prevents the teammates from becoming anonymous data points while keeping the interaction grounded in actual differences.

7. Iteration 1 - Shaping the band interface

Once the band metaphor was selected, the interface needed a clear hierarchy. The stage became the primary interaction area.

SKETCH 7
Personality Visuals



Simple round characters and small accessories added to show personality.

SKETCH 8
Synchronization



Different visual state for synchronization – drifting, syncing and in sync.

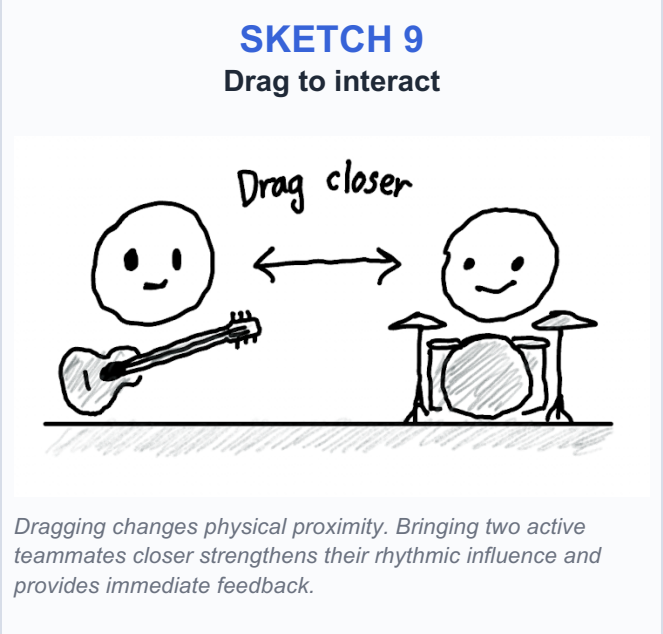
7.1 Why synchronization was added

The first band layout still risked becoming four independent musicians. We therefore added rhythmic synchronization so that activating a second or third person would create a new relationship, not simply add another sound. Similarity influences how strongly teammates pull toward a shared rhythm, so the application reveals something about the pair rather than only about each individual.

Visual connections reinforce the same idea. A less stable relationship appears more irregular, while a synchronized relationship appears more stable. Using both animation and short state labels reduces the amount of explanation required in the interface.

8. Iteration 2 - Adding interactive team dynamics

Synchronization improved the representation of relationships, but the user was still mostly observing. To make the team dynamic something the user could explore, we added dragging.



Dragging changes physical proximity. Bringing two active teammates closer strengthens their rhythmic influence and provides immediate feedback.

8.1 Why proximity matters

Dragging does not change personality similarity; the teammates remain the same people. Instead, proximity represents the strength of the current interaction. This distinction became central to the final concept: personality similarity represents who the teammates are, while distance represents how strongly they are interacting in the moment.

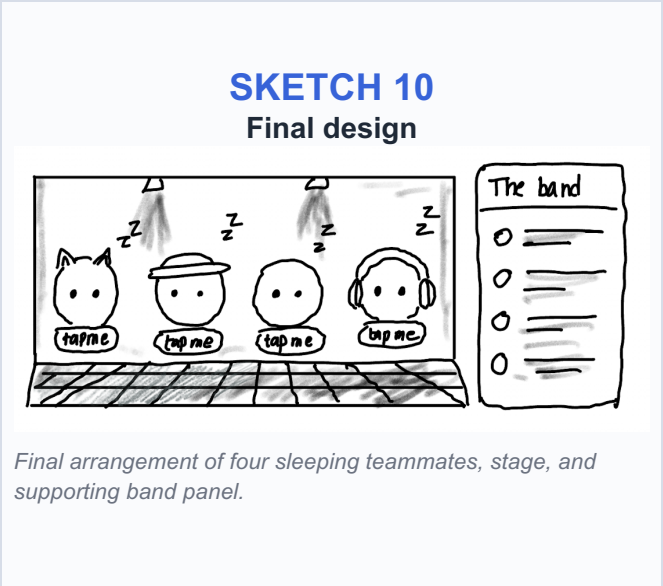
This also gives the user a meaningful experiment. A pair that does not synchronize naturally can be moved closer together to see whether stronger interaction helps them coordinate.

8.2 Evaluation of the working prototype

We evaluated the working design through criteria-based comparison, repeated team walkthroughs, and a usability heuristic review. The heuristic review focused on visibility of system status, user control, consistency, recognition rather than recall, and aesthetic/minimalist design. This was appropriate because the main design risk was not whether the code could run, but whether a first-time user could understand what activation, drifting, synchronization, and dragging meant.

Observed design issue	Refinement	Why the change mattered
The side panel became text-heavy as more personality information was added.	Move detailed explanations into secondary/collapsible content and keep the main state concise.	The stage remains the primary source of meaning; text confirms rather than dominates.
Dragging could feel disconnected from the result if the user had to wait for synchronization.	Strengthen immediate visual/status feedback during drag.	Improves visibility of system status and makes cause-and-effect clearer.
Pair relationships were difficult to perceive from sound alone.	Use relationship lines and state labels in addition to audio.	Supports recognition and makes the interaction understandable even without relying only on sound.
Utility controls could compete with teammate buttons.	Keep Reset and Sound visually secondary.	Preserves the assignment focus on one teammate button per member.

9. Final design sketch - Planned interface



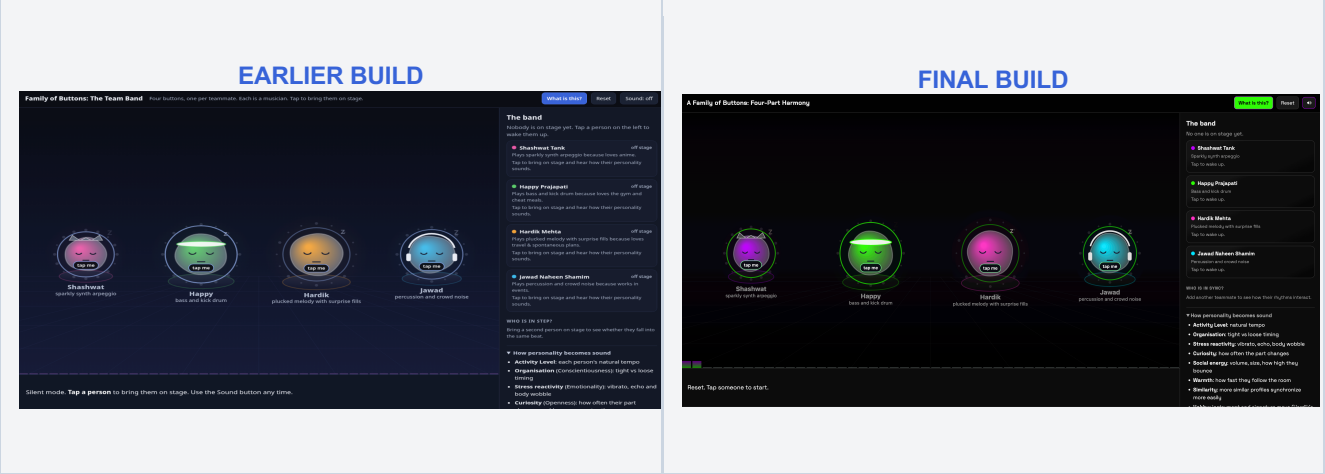
Final arrangement of four sleeping teammates, stage, and supporting band panel.

The final design sketch summarizes the interface direction established through the preceding iterations. It combines the four sleeping teammate buttons, shared stage, and supporting band panel into one planned layout.

10. Implementation refinements - From sketch to final build

After establishing the final layout direction, we continued refining the working implementation rather than treating the first functional prototype as the final design. Repeated walkthroughs showed that the interface could communicate the concept more clearly through a stronger visual identity, less text, more precise relationship feedback, and shorter instructions. The changes below connect the evaluation findings to specific implementation decisions.

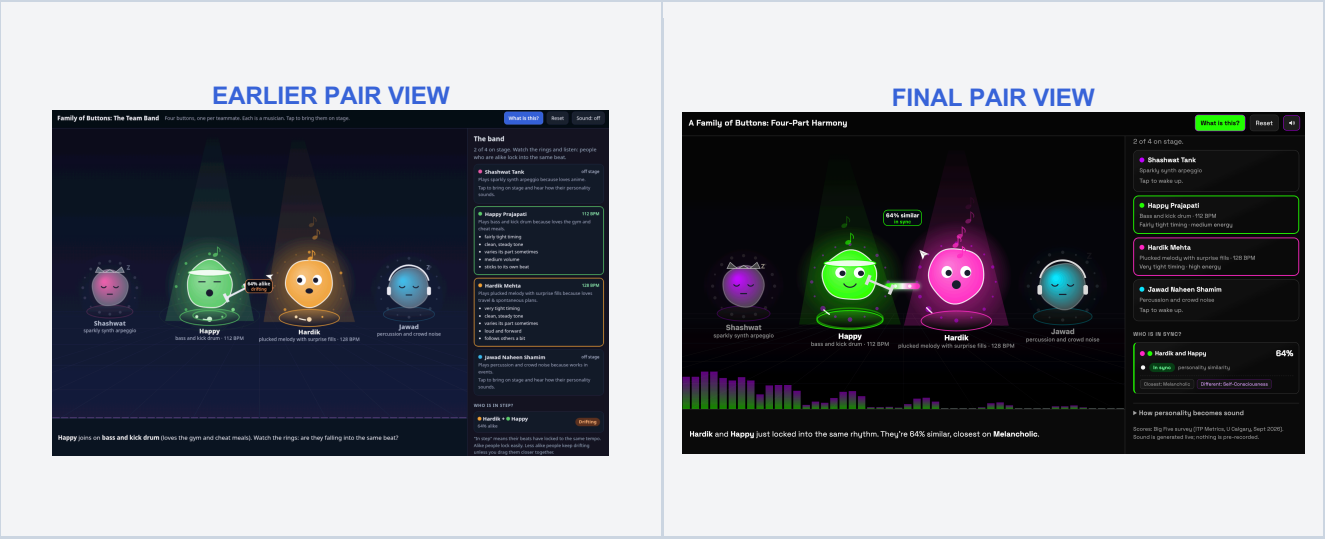
10.1 Colour and visual identity



Before and after comparison of the colour palette, header, and overall visual direction.

The earlier prototype used a dark blue interface and relatively muted teammate colours. During refinement, we used Figma's Color Combinations resource, the Neon Noir direction, as a reference for a higher-contrast concert-like aesthetic. The final build uses a near-black stage with neon purple, green, pink, and cyan accents. This made each teammate easier to distinguish against the background and made the application feel more like a live digital performance. We also applied colour consistently to interaction states: green supports the "In sync" state, purple supports "Drifting," and neutral grey is used while a pair is waiting. The change therefore supported both visual identity and system-status feedback.

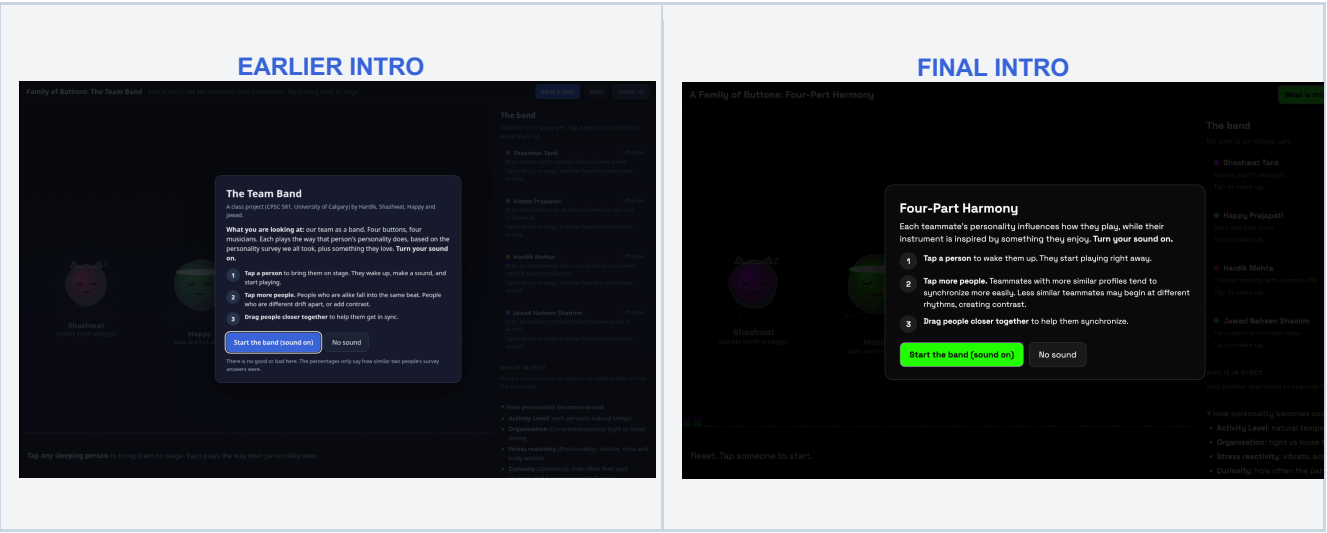
10.2 Information hierarchy and relationship feedback



Before and after comparison of the side panel and pair relationship feedback.

The first implementation placed several trait descriptions inside every active teammate card. Although informative, this made the side panel dense and competed with the stage. The final implementation reduces each active card to the teammate's musical role, BPM, and a short timing/energy description, while the detailed personality-to-sound mapping remains available in the collapsible section. The pair display was also redesigned. Instead of showing only an unexplained "% alike" value and synchronization state, it now labels the value as personality similarity and adds the personality facet on which the pair is closest and the facet on which they differ most. These changes make the panel easier to scan while giving the similarity percentage enough context to be meaningful.

10.3 Wording and control refinement



Wording refinement in the onboarding instructions, showing a move from broad claims to shorter and more precise language.

We changed several phrases after reviewing how a first-time user might interpret them. "People who are alike fall into the same beat" was changed to "teammates with more similar profiles tend to synchronize more easily." The new wording is more precise because the displayed percentage refers specifically to personality-profile similarity, and synchronization is also influenced by the interaction model rather than being guaranteed by similarity alone. Likewise, "% alike" became "personality similarity," and "in step" became "In sync," which better matches the music metaphor. "Bring them on stage" was shortened to "wake them up" because the sleeping character state already communicates what will happen visually. The header was also simplified; the repeated subtitle was removed, and the text-based sound control became a compact icon control. This keeps utility controls secondary to the four teammate buttons.

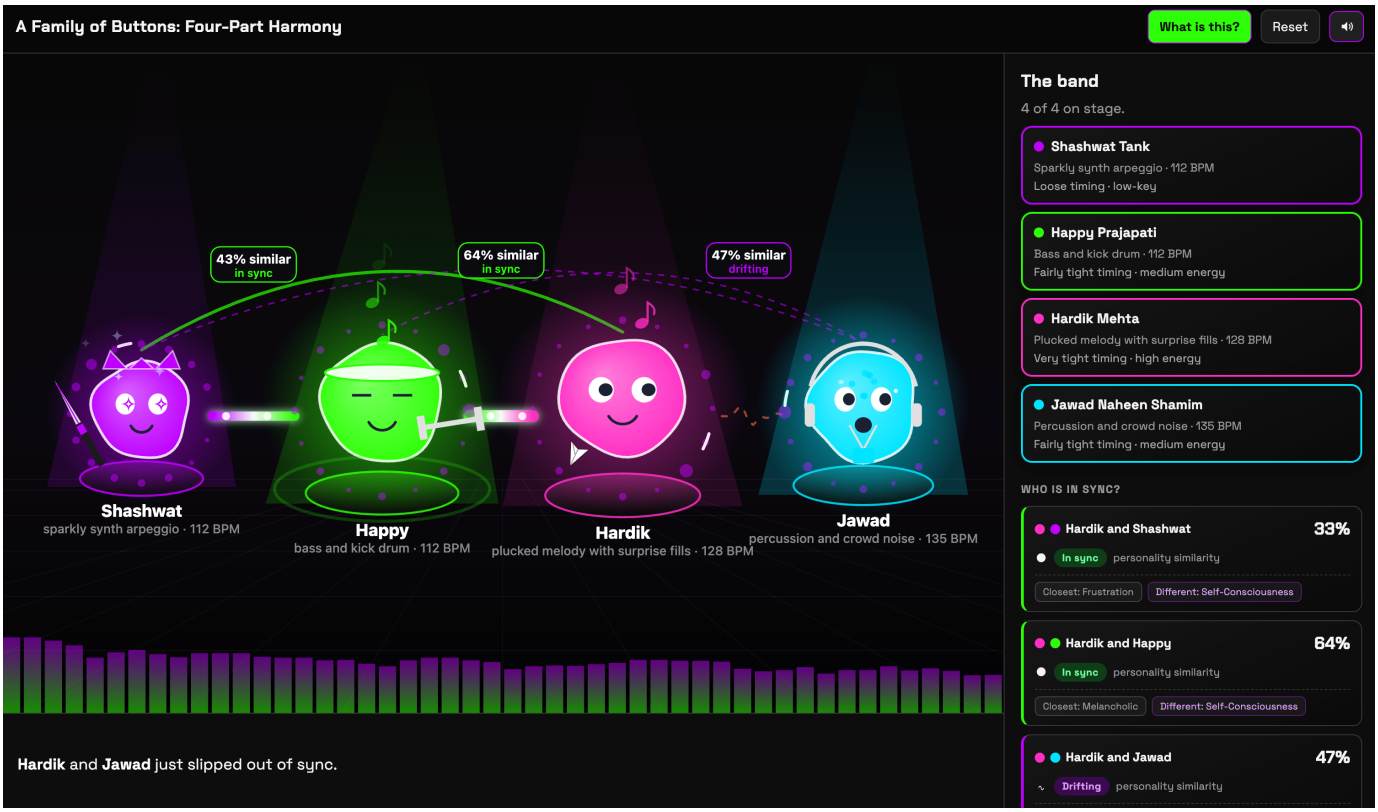
10.4 Coding changes supporting the refinements

Implementation change	Why it was made
Long active-card trait lists were replaced with a compact timing/energy summary.	Reduces visual density so the stage remains the primary interaction area.
Pair cards now generate “Closest” and “Different” facet tags in addition to the similarity percentage.	Explains what contributes to a pair relationship instead of presenting a percentage with no context.
Relationship states were standardized as Waiting, Drifting, and In sync and reinforced with consistent state colours.	Improves visibility of system status and makes pair behavior easier to recognize at a glance.
The sound control became an icon button with an ARIA label, and repeated header text was removed.	Keeps utility controls visually secondary while preserving a basic accessibility cue.

Together, these refinements turned implementation feedback into visible design decisions: the final version communicates more through the characters, state changes, and compact relationship cards, while explanatory text remains available when the user needs it.

11. Final application

The final application presents four circular teammate buttons on a shared stage. Inactive characters appear asleep and provide a clear "tap me" cue. Activating a teammate wakes the character, introduces their musical role, and begins personality-driven movement and sound. Activating additional teammates reveals pair relationships and team behavior.



Final application. The stage is visually dominant; the right panel provides supporting information about the band and personality to sound mapping.

11.1 Final design rationale

Design problem	Final response
Early concepts represented individuals but not the team.	Use one shared band in which every teammate contributes a distinct musical role.
Hobbies alone were too literal.	Use hobbies for recognizable identity and personality data for deeper behavior.
Four musicians could still behave independently.	Add pair synchronization so combinations reveal team dynamics.
Users could observe relationships but not influence them.	Add dragging so proximity changes interaction strength.
Relationship behavior could be hard to interpret.	Use visual connection states plus short explanatory feedback.
Explanations risked becoming too text-heavy.	Make stage behavior primary and move detail into supporting/collapsible content.

12. Appropriateness of methods and tools

The methods and tools were selected according to the stage of the design process. Early methods favored speed and breadth; later methods favored comparison, implementation, evaluation, and refinement.

Method / tool	Why it was appropriate	How it informed a decision
Low-fidelity iPad sketching	Fast exploration before implementation.	Revealed that interest-based individual concepts were recognizable but weak at showing team relationships.
Divergent ideation	Encouraged multiple metaphors rather than locking into the first idea.	Produced individual concepts, a shared environment, and sound-based directions.
Criteria-based convergence	Connected concept selection to the assignment rather than personal preference.	The band was selected because it supported both individual identity and team dynamics.
Big Five Personality (ITP Survey)	Provided structured, non-arbitrary differences between teammates.	Traits were mapped to tempo, timing, variation, movement and responsiveness.
Iterative prototyping	Allowed interaction problems to be found after implementation.	Led to synchronization, dragging, stronger feedback and reduced text.
Usability heuristic review	Provided a systematic way to examine clarity and feedback.	Highlighted system-status visibility, user control, consistency and minimalist design.
HTML / CSS / JavaScript	Supported rapid browser-based iteration with no installation required for users.	Enabled one-file deployment and direct manipulation of teammate buttons.
Canvas and Web Audio API	Supported custom animation and generated sound tied to live state.	Made personality and team dynamics sensory rather than purely textual.
Figma Color Combinations	https://www.figma.com/resource-library/color-combinations/ Supported color combination (51, Neon Noir).	We assigned each teammate a separate accent color so users could distinguish characters and relationship indicators quickly.
GitHub	Supports versioning, collaboration and web deployment.	Keeps implementation changes traceable and makes the final app easy to host.
Microsoft Word	Useful for document collaboration, usability review and wording.	Microsoft Word helped us organize the report visually and chronologically, making the progression from early ideas to the final design clear.

13. Known incomplete, broken, or missing functionality

At the time of preparing this report, no core feature is intentionally incomplete in the current build. The application supports activating and removing teammates, personality-driven audio and animation, pair relationship feedback, dragging, reset, and sound control.

Known limitations: the Drifting / In Sync state can briefly fluctuate during the first few seconds after two teammates join the stage while the synchronization model stabilizes. Character names and instrument labels may also overlap when two characters are dragged very close together, and some accessibility limitations remain because the application has not undergone a complete accessibility review.

14. Generative AI disclosure

Generative AI was used during this project and is disclosed because the assignment requires the team to state which tools were used, how they were used, and how the outputs were incorporated.

Tool	Use in the project
Claude (Anthropic)	Used to assist with code implementation based on the team's specifications and architectural decisions. The team defined the required features, personality-to-audio mappings, interaction model, audio direction, and visual direction, while Claude assisted in translating these requirements into working HTML, CSS, and JavaScript.
ChatGPT (OpenAI)	Used for rubric review, usability and heuristic feedback, exploration and visualization of sketch ideas, wording refinement, code/interface feedback, report organization, and formatting of the final documentation. The team reviewed and selected which suggestions to incorporate.

AI-generated material was treated as support rather than an automatic final answer. The team selected the concepts to keep, reviewed the application behavior, adapted the interface, and remains responsible for the submitted work.

14.1 Examples of prompts used

- **Prompt 1 - Initial Build (Claude)**
  Build a single self-contained HTML/CSS/JS web app called 'Four-Part Harmony.' It's 4-character buttons, one per teammate (Hardik, Shashwat, Happy, Jawad), each generating live synthesized audio (Web Audio API oscillators/synths) based on personality percentiles. Map specific traits to specific audio parameters explicitly: Activity Level → tempo, Conscientiousness → timing tightness, Extraversion → volume, Emotionality → vibrato, Openness → pattern-change probability, Agreeableness → sync coupling strength (Kuramoto-oscillator phase model). Tapping a character wakes it up and it starts playing its own instrument (each person gets a distinct synthesized voice plucked synth, bass/kick, percussion/crowd noise, arpeggio/bells). Tapping multiple shows pairwise sync/drift as animated linking lines between them, with a closest/furthest trait tag per pair and a percentage similarity score. All four playing at once triggers a FULL BAND! state. Visual style: black background, neon-noir palette electric green, purple, cyan, magenta accents on near-black panels. Fonts: Chakra Petch for headings, Space Grotesk for body text, loaded via Google Fonts CDN.
- **Prompt 2 - Version 1 to Final (Claude)**
  Remove the tour thing the auto-playing 50-second guided tour and the 'Tap all four' shortcut are both cut from the header. Tap all four does nothing a person couldn't already do themselves in four taps, and the tour competes with the one interaction that actually matters tapping a character and hearing what happens. Keep the header to three controls: What is this? / Reset / Sound. The palette still looks washed out match the colors to the reference image exactly (black/neon-green/electric-purple/cyan/magenta).
- **Prompt 3 - Documentation Formatting (ChatGPT)**
  The content is done and final don't add, cut, or rewrite any of the substance, and don't touch the wording unless it's a typo or grammar error. I just need it reformatted and cleaned up for submission apply consistent headings/section numbering, fix inconsistent spacing, bullet styles, and heading levels.

15. Conclusion

Four Part Harmony developed from separate ideas about individual teammates into one interactive representation of the group. Divergent sketching exposed the limits of isolated personal concepts; convergence around the band metaphor connected individual personality to sound and movement; and later iterations added synchronization, dragging, clearer relationship feedback, a stronger visual identity, and more precise wording. The final application therefore combines recognizable likeness, personality-driven behavior, and interactive team dynamics while staying focused on the project goals of novelty, clarity, and reliable execution.